SIGNAL OFFICER ADVANCE COURSE | TOPIC 5.4

C4ISTAR Sensors Integration

Operationalizing the Sensor-to-Shooter Link across the Multi-Domain Battlespace

SENSOR **DECISION-MAKER** **SHOOTER**

Module 5: Advance C2 Communications

Primary reference: PAM 6-0001 — Command and Control, Communications, Computer, Intelligence, Surveillance, Target Acquisition, and Reconnaissance (26 Aug 2024)

INTRODUCTION | INTEREST

A Drone, a Cellphone, and 18 Hostages

Marawi, 2017

During the Marawi Crisis, government and Maute Group forces both fought “the drone war,” using commercial drones for ISR. In one operation, a drone was used to deliver a cellphone that helped locate and coordinate the rescue of 18 hostages.

Ask the class:

What had to go right — technically and organizationally — for that drone, that cellphone, and the rescue team to work together in time?

We will return to Marawi later in this session — including where the integration broke down.

INTRODUCTION | NEED

Why This Topic Matters to You



As a Signal Officer, you will own this loop

Integrated sensors are what let commanders maneuver forces to advantageous positions and attack enemy locations at the commander's call.

Integrated sensor data feeds the **Common Operational Picture (COP)**: situational awareness, focused intelligence, collaborative planning, mobility, efficient logistics, and accurate counterfire.

As future Signal Officers, you will be the ones integrating, testing, and sustaining the sensors that feed this loop — getting this wrong costs lives in contact.

Terminal Learning Objective

At the end of the session, the students will be able to **demonstrate proficiency in integrating C4ISTAR sensors** to enhance military operational capabilities in accordance with PAM 6-0001. **(C3)**
.

Enabling Learning Objectives:

1. **ELO 5.4.1** **Discuss** the Concept and Importance of C4ISTAR Sensors Integration. (C3)
2. **ELO 5.4.2** **Determine** Types and Capabilities of C4ISTAR Sensors. (C3)
3. **ELO 5.4.3** **Discuss** C4ISTAR Requirements and Operational Context. (C3)
4. **ELO 5.4.4** **Discuss** the Operational Scenarios for Sensor Integration. (C3)
5. **ELO 5.4.5** **Discuss** the Integration and Testing of C4ISTAR Sensors. (C3)

INTRODUCTION

References

PRIMARY REFERENCE

PAM 6-0001

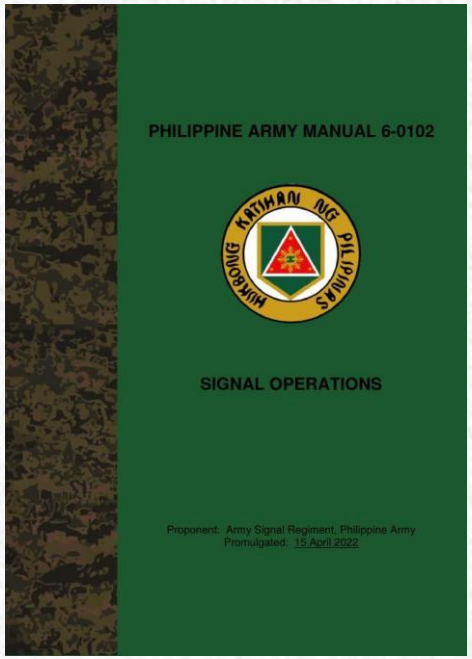
Command and Control, Communications, Computer, Intelligence, Surveillance, Target Acquisition, and Reconnaissance (Philippine Army)

Promulgated 16 Aug 2024

ALSO USED

Student Handout: C4ISTAR Sensors Integration

Session Plan: C4ISTAR Sensors Integration



PAM 6-0102

What is C4ISTAR?

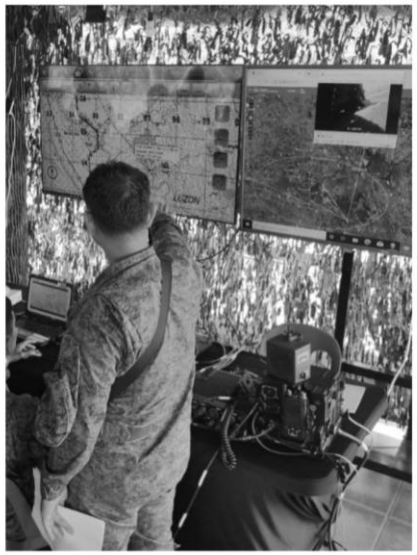
A System of Systems

C4ISTAR refers to a system of systems that integrates and shortens the **Sensors, Decision-Makers, and Shooters Loop** to deploy forces and resources in support of mission accomplishment — giving commanders at all levels an accurate, timely, and correct picture of the battle space.

PAM 6-0001 Preface, ¶0201



SEE FARTHER



DECIDE FASTER



PROSECUTE WITH PRECISION

The Sensor–Decision-Maker–Shooter Loop

Sensor

Sees — detects and collects data from the environment.

Decision-Maker

Thinks — analyzes the data, assesses the situation, decides.

Shooter

Acts — executes the decision.

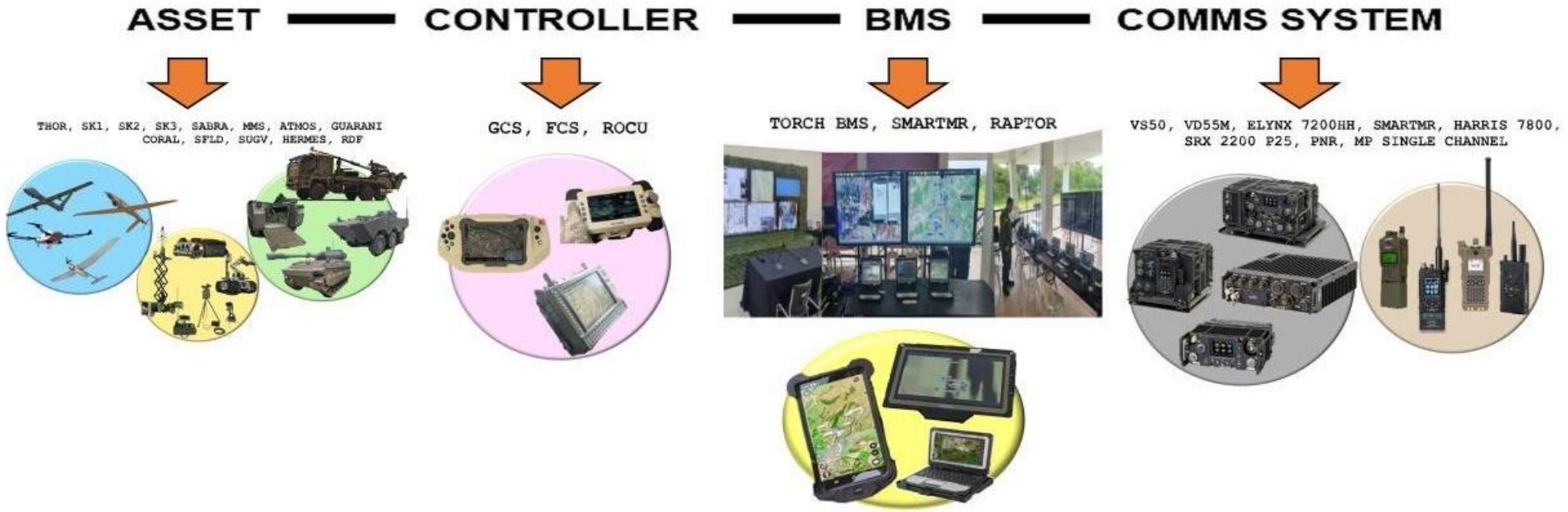
Where Sensor Integration Fits

Sensor integration is what keeps that first link — seeing — fast, accurate, and connected to everything downstream of it.

The Sensor–Decision-Maker–Shooter Loop

Sensor/Weapon System

❑ Basic configuration of sensor or weapon system is as follows:



- The Controller enables the operator to manipulate the asset as his extension
- The BMS is where SA&COP is achieved
- The Comms Sys provides Command & Control

Why Sensor Integration Matters

- Provides the **situational awareness** that allows Combined Arms Operations to be conducted — letting commanders maneuver forces and attack enemy locations at their call.
- Feeds the **Common Operational Picture (COP)**: battlespace situational awareness, focused intelligence, collaborative planning, mobility and flexibility, efficient logistics, and accurate counterfire.
- **Integration** is combining multiple applications so different software, systems, and processes function as one uninterrupted system — sensor integration applies this to the PA's sensor inventory.

PAM 6-0001 ¶0211–0214

The Five PA C4ISTAR Sensor Types

Sensor	Capability
Sniper Fire Location Detector	Detects and locates the source of sniper fire in real time.
Thermal Imager	Portable, battery-operated device for long-range observation by recon teams or forward observers.
Ground-Based Radio Direction Finding Systems	Finds the direction of a radio source; two or more measurements locate an unknown transmitter or vehicle.
Unmanned Aerial Systems (UAS)	Light air vehicles carrying sensor payloads, used for remote sensing and mapping missions.
Small Unmanned Ground Vehicle (SUGV)	Modular platform that plugs in multiple payloads, reduces soldiers' exposure to hazards, and delivers squad-level SA.

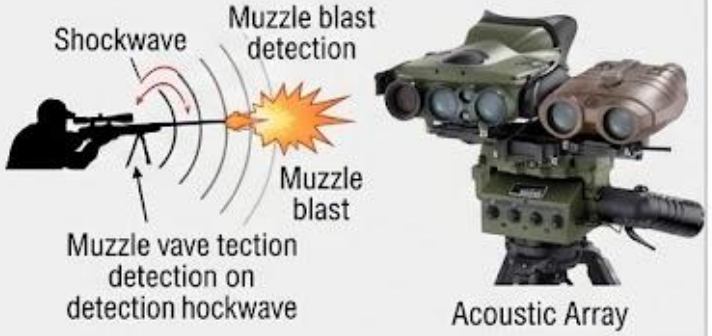
PAM 6-0001 ¶0353–0358

The Five PA C4ISTAR Sensor Types

ACOUSTIC & UNMANNED GROUND PLATFORMS

○ CAPABILITIES

Sniper Fire Location Detectors





Small Unmanned Ground Vehicles (SUGV)

Dangerous Reconnaissance



Dangeron camera inspecing a suspicious objects

EOD Clearing Operations



EOD cleard operations, manipulang a potential explosive

○ CONSTRAINTS & VULNERABILITIES

Acoustic Sensors

Constraints
- Multipath reflections in urban areas
- Limited range in high noise, high data infection

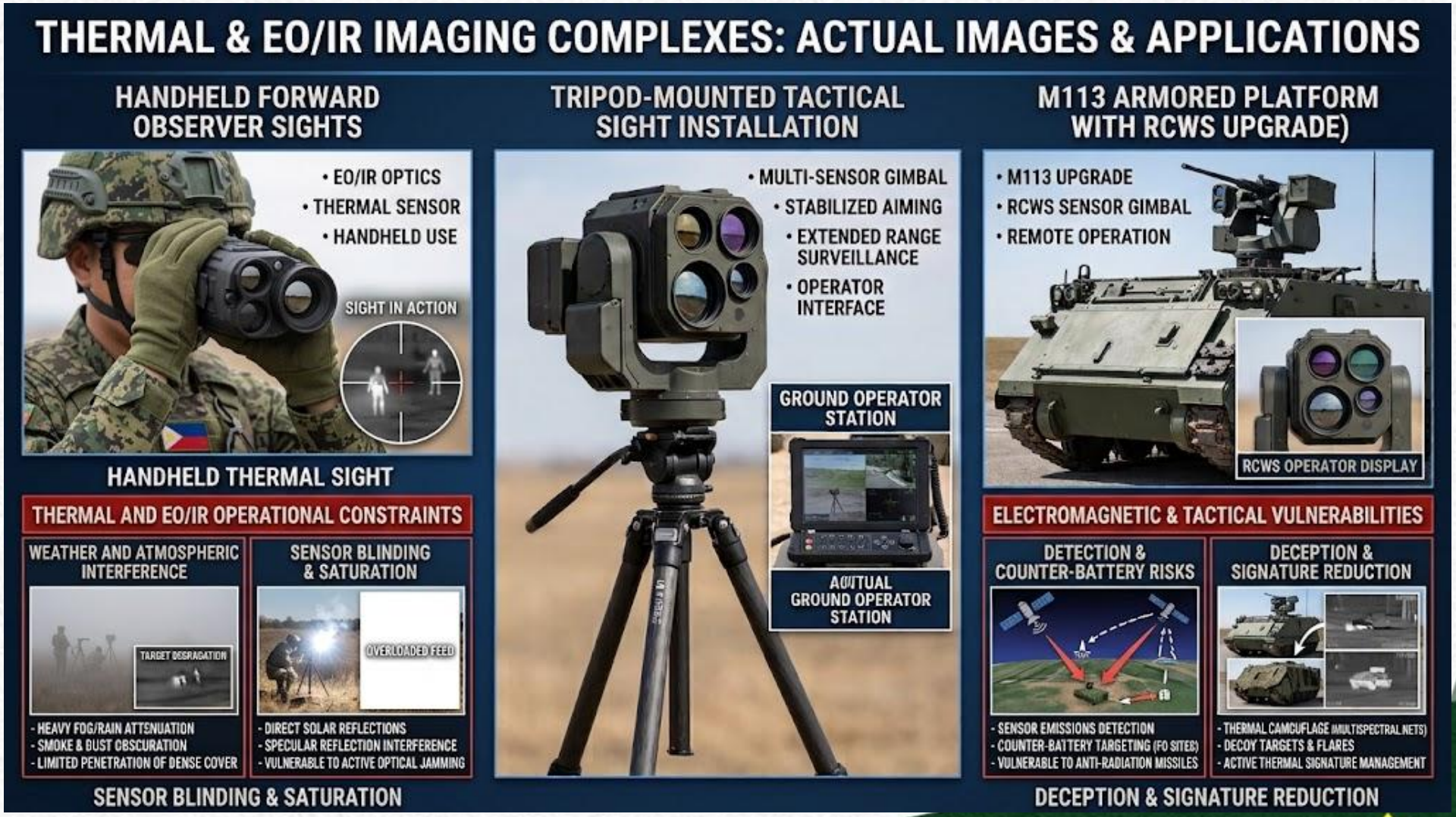
Vulnerabilities
- Weather (wind/rain) and watercloogs
- Acoustic decoys to decoys

SUGV

Constraints
- Battery endurance in battery endurance
- Communication link in tunnels/buildings

Vulnerabilities
- RF jamming
- Physical capture & physical capture / damage

The Five PA C4ISTAR Sensor Types



The Five PA C4ISTAR Sensor Types

SIGNALS INTELLIGENCE (SIGINT) & RADIO DIRECTION FINDING (RDF): CAPABILITIES & CONSTRAINTS

○ CAPABILITIES

1. RF Interception & Spectrum Scanning



2. Emitter Classification

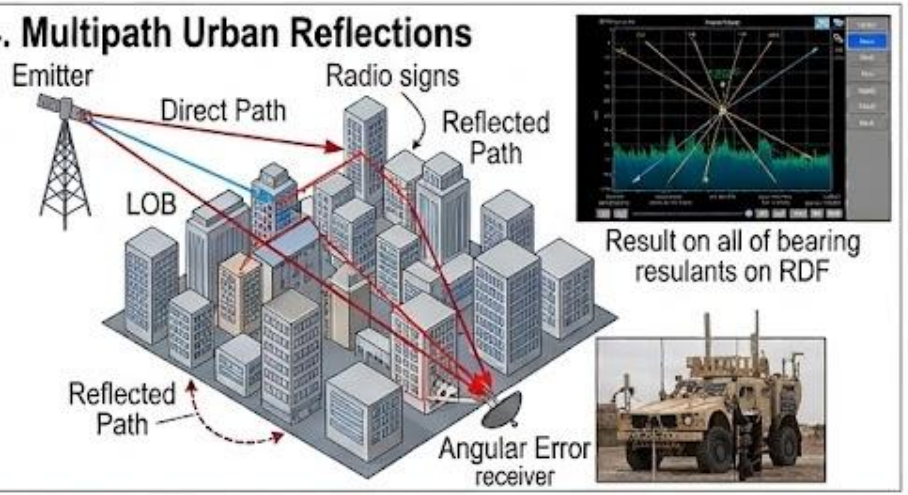


3. Line-of-Bearing & Triangulation (DOA)

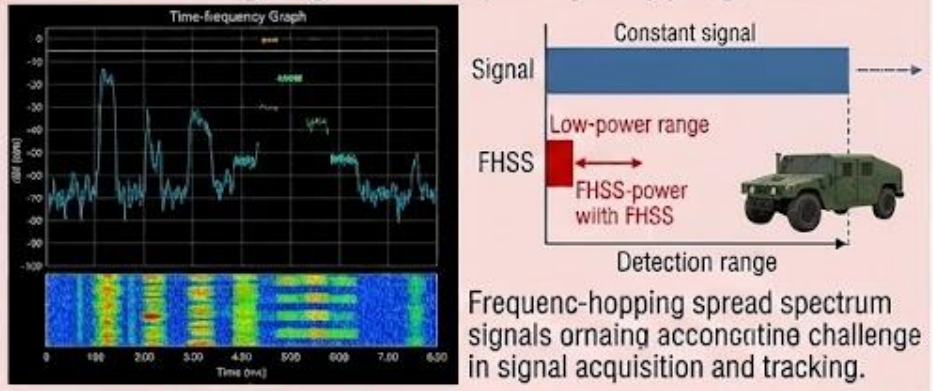


○ CONSTRAINTS

4. Multipath Urban Reflections

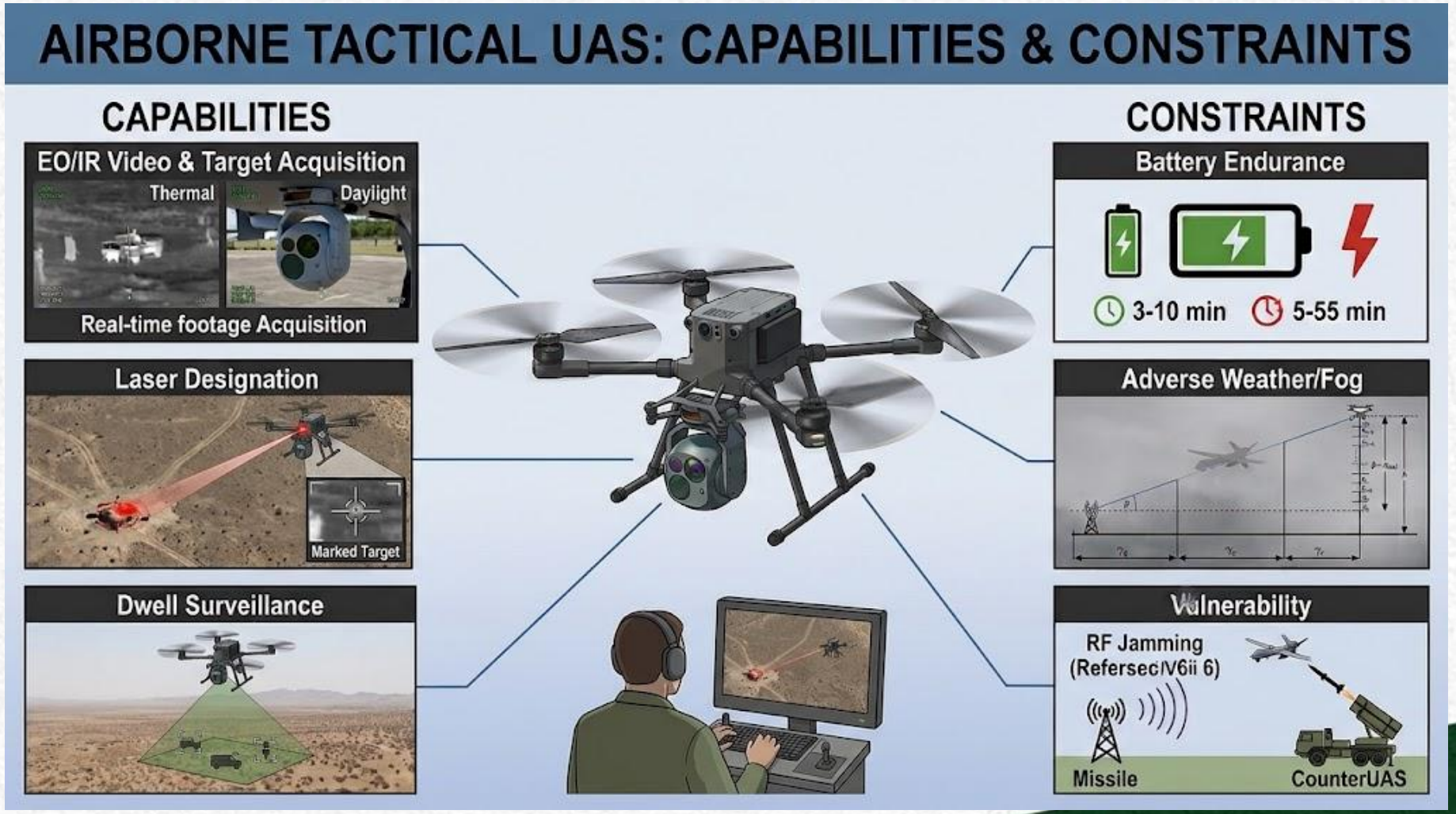


5. Limited Range against Frequency-Hopping Emitters



Frequenc-hopping spread spectrum signals ornaing acconcatine challenge in signal acquisition and tracking.

The Five PA C4ISTAR Sensor Types



Grouping the Sensors by Job

Acoustic / RF Detection

Sniper Fire Location Detector, Ground-Based Radio Direction Finding Systems

Optical Observation

Thermal Imager

Mobile / Remote Platforms

UAS, SUGV — carry other sensors to where fixed ones can't reach

How Sensors Connect Back

Connected through **radio frequency** from the ground control system to the command post, and through the **PA Network (PANET)** back to Headquarters.

PAM 6-0001 ¶0359

Points of Interoperability

- **Interoperability Points (IOP):** national networks can use whatever technology they choose internally — only the connection points between networks need a defined, standardized protocol stack.
- **NATO Interoperability Standards and Profiles (NISP):** prescribe the technical standards needed to achieve interoperability of people, processes, technology, and communication information systems across an alliance.
- **Interface Control Document (ICD):** the technical document describing a system's interfaces and communication rules — tells integrators how to connect sensors to the rest of C4ISTAR and confirm compatibility.
- **Open Architecture:** lets new sensors and capabilities be added onto the existing system as "add-ons," keeping integration and future upgrades cheaper and faster.

PAM 6-0001 ¶0319–0327

Media of Communication a Sensor May Ride On

Fixed communication

- The PA's backbone for voice and data
- **BONTEX:** telephone exchange up to GHQ
- **PANET:** secured, managed intranet
- Extended to brigades through DBTOCS

Tactical communication

- Voice is the bedrock of tactical communication
- **OTM** (on-the-move) and **OTH** (on-the-halt) devices
- HF, VHF and UHF radios; combat net radio (CNR) using ROIP and ad-hoc networks
- Non-military radios complement, for other agencies

Special purpose communication

- Commercial devices for routine or administrative traffic
- Videoconferencing, cellular and satellite phones
- Use with **utmost prudence**, guided by COMSEC policy

PAM 6-0001 ¶0325a–c

Constraints Every Integration Effort Must Account For

Challenge	What It Means for Sensor Integration
Interoperability	Different vendors mean different protocols/standards — seamless data sharing is hard by default.
Cybersecurity	More interconnected sensors mean a bigger attack surface for cyber-attacks and espionage.
Complexity	Many components, networks, and technologies make reliability and training a constant task.
Cost	Developing, deploying, and maintaining sensor capability is expensive and must fit budget constraints.
Legacy Integration	Older systems may not be compatible with new sensors — backward compatibility must be engineered in.
Data Overload	ISR sources can generate more data than operators can process without good data management.
Environmental Factors	Urban, desert, mountainous terrain, extreme temperatures and EMI all affect sensor reliability.
Human Factors	Operator training, situational awareness, and human error remain critical even with automation.

PAM 6-0001 ¶0230–0237

Three Operational Contexts

Operational Context	What Sensor Integration Must Deliver
Combined Arms Operations (CAO)	A secure, robust, reliable network linking leaders and soldiers across a dispersed battlespace, with sensor-fed SA enabling quick commander decisions.
Joint Operations	PA sensors feed the ground-defense portion of the AFP Joint C4ISTAR, contributing to a COP shared across Major Services.
Coalition Operations	Interoperability with partner-nation systems on different standards — exercised through UN peacekeeping and exercises such as Balikatan.

PAM 6-0001 ¶0210–0229

Case in Point: The Marawi Crisis (2017)

The Drone War, and Where Integration Broke

Both government and Maute Group forces used **commercial drones** for ISR — including a drone that delivered a cellphone, helping locate and coordinate the rescue of 18 hostages.

The campaign also exposed a hard integration gap: the existing radio network (SMVRS) could carry **voice only**, not the data a Blue Force Tracking (BFT) system needed — forcing a second SMVRS to be stood up for data. A demo BFT system pairing a Harris SA recorder with Esri's ArcGIS GeoEvent Processor broke down when its **software license expired mid-operation.**

PAM 6-0001 ¶143–0164

The lesson: sensor integration has to be tested and sustainable before it is needed in contact, not improvised during it.

Marawi Siege (2017): Operational Lesson Learned



Physical Courier Delays

Lack of drone-to-TOC datalinks forced physical transit of SD cards under fire.



Voice-Only Tactical Networks

Repeaters lacked data capability; teams crawled under fire to reconfigure radios for BFT.



Commercial Software Failure

Expired commercial server demo licenses blinded command tracking mid-campaign.

The Marawi Liberation (2017): Birth of the C4ISTAR Horizon 2



Seeing in the Dark
Thermal Imaging Binoculars



Eyes in the Sky
Unmanned Aerial System



Secured, Interconnected Comms
Combat Network Radio System



Locating Snipers
Sniper Fire Locator System



Sifting Through Rubbles
Small Unmanned Ground Vehicle



Protect fm IED
Man Portable Jammer



Locate Enemy RF
Radio Direction Finder

The PA C4ISTAR (2025): 8 Years After the Marawi Liberation

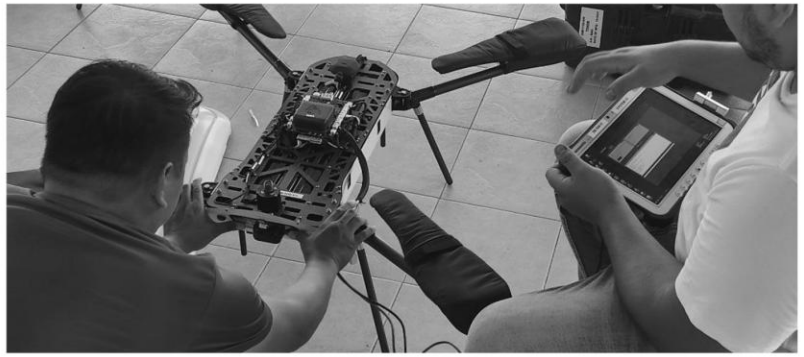
Integrated Sensors, Trained Personnel

- Full integration of 5 Lots using the Elynx MANET Radios through a BMS
- Achieved situation awareness (SA) and common operating picture (COP)
- Full knowledge transfer of setup, configuration and operation of Elynx Radios











The PA C4ISTAR (2025): 8 Years After the Marawi Liberation

Arrival of New Weapon Systems: Closing the Loop of Sensor, Shooter, & Decision Maker

- DND issued a memorandum for H2 CNRS Project to provide Elynx MANET Radio for ATMOS and MMS.
- PA only issued MANET Radios for 10 ATMOS and 5 MMS.
- All issued radios were configured and integrated with the H2 Sensors for target acquisition
- The Armor Division and Artillery Regiment awaits the issuance of lacking Elynx Radios







The PA C4ISTAR (2025): 8 Years After the Marawi Liberation

Integration of the PAF Hermes 900

- Shared Video with Position Location Information of PAF UAS with the PA Operation Center
- Voice Communication with the GCS Pilot
- Provide secured network to share video







Integration of PA C4ISTAR System

Setup C4ISTAR Laboratory for Horizon 3 CNRS MANET Proof of Concept

The PA established the C4ISTAR Laboratory Concept to build a benchmark and ensure new capabilities seamlessly integrate with existing systems.









Who Integrates the Sensor

Army Signal Regiment (ASR)

The C4ISTAR Capability Champion — responsible for the Installation, Operation, and Maintenance (IOM) of core C4 systems.

Army Intelligence Regiment (AIR)

Responsible for the IOM of ISTAR core systems — the sensors themselves.

How They Coordinate

ASR handles installation, repair, and maintenance of C4ISTAR hardware and software **in coordination with AIR**; end-user hardware operation and preventive maintenance stay with the using unit. **Technical experts** keep data collected, processed, and disseminated accurately to decision-makers.

PAM 6-0001 ¶0245, ¶0351

Integration vs. Interoperability



Integration

The process of combining multiple applications so different software, systems, and processes function as one uninterrupted system.



Interoperability

The ability of architecturally different software, systems, and processes to speak the same language through real-time data exchange.

Integration is getting the sensor talking to the rest of C4ISTAR; interoperability is proving it keeps talking under real conditions.

Testing and Design Requirements

- **Regular operational testing** must identify and address vulnerabilities or malfunctions before a sensor is trusted in the field.
- **Redundancy** must be built in so the failure of one piece of equipment does not compromise the entire system.
- **EMI and radio-frequency shielding** must protect the sensor against electronic warfare and other electromagnetic threats, including EMP.
- **Modular design** allows components to be replaced or upgraded without affecting the whole system; compatibility stays a design requirement, not an afterthought.
- The PA's **depot-level and level-4 repair facilities** assess next-generation technologies and evaluate/validate technical systems — the institutional home for ongoing sensor testing.

PAM 6-0001 ¶0406–0407, ¶04015

Key Takeaways

- C4ISTAR runs on the **Sensor–Decision-Maker–Shooter loop** — integration keeps the first link fast, accurate, and connected.
- The PA fields **five sensor types**: Sniper Fire Location Detector, Thermal Imager, Ground-Based RDF, UAS, and SUGV — connected via RF to the CP and PANET to HQ.
- Integration must satisfy the **operational environment** (physical and non-physical domains) and achieve interoperability through standards like NISP and documents like the ICD.
- The **right sensor and integration strategy change with the operation** — Combined Arms, Joint, or Coalition — and Marawi shows why integration must be proven before contact.
- **ASR** installs and maintains core C4 systems; **AIR** owns ISTAR (sensor) systems — they coordinate on IOM.
- **Integration ≠ interoperability** — interoperability is proven only through testing, redundancy, shielding, and modular design.

EXPERIMENT | GROUP PLANNING EXERCISE

Simulated Operational Scenario

1. **Littoral Patrol**



2. **Urban Operations**



3. **Humanitarian Assistance and Disaster Response**



Present how C4ISTAR sensors can be integrated to support mission accomplishment.

EXPERIMENT | GROUP PLANNING EXERCISE

Presentation Requirements

Part	Focus	What to Cover
Part I	Introduction	Definition of C4ISTAR Sensors Integration; importance in modern military operations
Part II	Sensor Analysis	Identify and explain at least five (3) sensors applicable to the chosen scenario
Part III	Integration Strategy	How sensors will be integrated; data flow; C2 support; SA enhancement
Part IV	Operational Application	Expected mission execution; advantages; limitations; recommendations
Part V	Conclusion	Summary of key findings and lessons learned

Deliverables: Oral Presentation (10–15 minutes) • PowerPoint Presentation (8–10 slides)

REFLECT

Before You Present…

- Finalize and rehearse your group's presentation.
- Check that every slide cites PAM 6-0001 and the Student Handout where it should.
- Raise any questions, concerns, or clarifications with the instructor before Evaluation begins.

EVALUATION

Group Presentation Rubric

Criteria	Weight
Content Accuracy	25%
Application to Scenario	20%
Critical Analysis	15%
Organization and Structure	10%
Visual Aids	10%
Team Participation	10%
Delivery and Communication Skills	10%

Students must obtain at least 75% of the rubric criteria to pass the subject.

SIGNAL OFFICER ADVANCE COURSE | TOPIC 5.4

Sensors See. Networks Decide. Shooters Act.

As future Signal Officers, your mission is making sure that loop never breaks — from forward sensor to kinetic effect.

NEXT TOPIC: 5.5 — C4ISTAR Weapon System Integration